

# Quality-Aware LLM Data Augmentation for Arabic Dialectal NER

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## Abstract

Named Entity Recognition for dialectal Arabic remains severely constrained by data scarcity, and LLM-based augmentation is the natural remedy. We study LLM-based augmentation for the Lebanese and Palestinian sub-corpora of Wojood 2024, using LLaMA-3.1-8B-Instruct as a generation backbone, and compare three strategies (entity-level, context-level, combined) across two  $k$ -shot sampling paradigms. The proposed multi-dimensional pipeline yields a per-sentence composite score  $Q(s)$  that exposes a sharp dialectal-authenticity gap between entity- and context-level rewriting (DA 0.92 vs. 0.51). Despite lower register fidelity, context augmentation with curriculum training delivers the most consistent gains (Curras F1 +5.70 pp); entity augmentation lifts Lebanese F1 from a zero baseline to 50.00 on Lebanese-only training, a figure the 85-sentence test set makes coarse. Ablation, weight-sensitivity, threshold sweep, a random-selection control, and a second LLM backbone show that the dialectal-authenticity dimension carries most of  $Q(s)$ 's protective effect on Lebanese and that generator choice is itself a first-class register-preservation lever.

## 1 Introduction

Named Entity Recognition is a foundational NLP task, yet its performance degrades dramatically on dialectal Arabic text. Arabic combines a morphologically rich writing system, high lexical ambiguity, and a wide spectrum of dialectal varieties (Lebanese, Palestinian, Egyptian, Gulf, Maghrebi), each with non-standard orthography, regional vocabulary, and endemic code-switching. Modern Standard Arabic NER benefits from large annotated corpora and strong pre-trained encoders (Antoun et al., 2020; Inoue et al., 2021; Abdul-Mageed

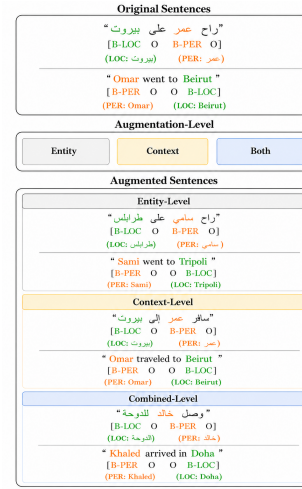


Figure 1: Overview of the three augmentation strategies (entity-level, context-level, and combined).

et al., 2021), but dialectal variants remain severely under-resourced.

The Wojood 2024 benchmark (Jarrar et al., 2024) is the most comprehensive Arabic NER dataset to date, covering 13 sub-corpora and 21 fine-grained entity types. Its dialectal content concentrates in two sub-corpora, Lebanese (298 training sentences, 1.3%) and Curras/Palestinian (3,660 sentences, 15.8%), while 82.9% of the data is predominantly MSA. This imbalance produces a striking disparity: a model trained on Lebanese data alone reaches **near-zero F1**, whereas the full benchmark yields 86.48 F1.

LLMs have been applied to NER augmentation with success in resource-rich settings (Ye et al., 2024; Wang et al., 2025), yet no prior study has investigated LLM augmentation quality specifically for Arabic dialectal NER. We argue that the dialectal setting introduces a qualitatively different failure mode from those studied in English- or MSA-only augmentation: *cross-register drift*. Multilin-

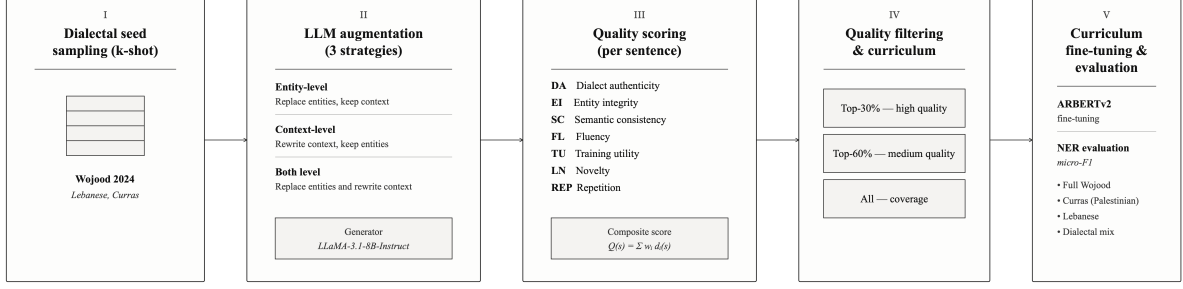


Figure 2: End-to-end pipeline. Stage I restricts sampling to the dialectal sub-corpus to preserve register; Stage II generates augmented sentences under three strategies; Stage III evaluates each sentence and produces  $Q(s)$ ; Stage IV trains ARBERTv2 using quality curriculum tiers.

gual LLMs carry a strong MSA prior, so rewriting a Lebanese or Palestinian sentence silently produces a fluent MSA paraphrase that satisfies every generic quality signal while destroying the dialectal supervision the augmentation was meant to provide. The contribution of this work is therefore not a new filter, but a *dialect-aware* characterisation of which quality dimensions actually matter for low-resource dialectal NER.

We make the following contributions:

1. A **three-level augmentation pipeline** for Arabic dialectal NER combining entity-level, context-level, and hybrid strategies with LLaMA-3.1-8B-Instruct on Wojoood 2024.
2. A **composite quality score**  $Q(s)$  with full ablation and weight-sensitivity analysis, identifying register fidelity as its dominant component.
3. Empirical evidence of a **dialect preservation gap**: entity-level augmentation preserves register by construction; context-level rewriting drifts toward MSA under prompts that do not require the seed register to be kept.
4. **Generator generalisation** across two backbones, revealing a register-dependent dialectal ceiling on Lebanese that makes generator selection a register-preservation design variable.

## 2 Related Work

**Arabic NER and dialectal resources.** Arabic NER has progressed from MSA corpora such as ANERCorp (Benajiba et al., 2007) and AQMAR (Mohit et al., 2012) to the multi-corpus Wojoood benchmark (Jarrar et al., 2022, 2024). Dialectal

performance remains systematically lower due to morphological divergence, regional lexicon, and inconsistent orthography. Dialect-aware pretraining (Shang et al., 2025; Alimi et al., 2025), multi-dialect instruction datasets such as PALM (Alwajih et al., 2025), and the BALSAM benchmark (AlMatham et al., 2025) all identify dialectal NER as a persistent bottleneck even for large models.

**LLM-based augmentation for NER.** Early Arabic-specific work used rule-based or label-aware perturbations (Ghazoui et al., 2026, 2025). These rule-based pipelines preserve register by design (no generation step) but produce only minor lexical variation.

More recently, LLMs have enabled two augmentation paradigms: *controlled rewriting*, combining entity substitution with context paraphrasing (Ye et al., 2024; Kang et al., 2024), and *fully generative* approaches synthesising BIO-tagged sentences from seed entities or bidirectional contexts (Dao et al., 2025; Wang et al., 2025). Both lines report consistent gains on English and high-resource benchmarks: LLM-DA (Ye et al., 2024) reports F1 improvements in the +1–3 pp range on CoNLL-2003 and OntoNotes; LM-Transplant (Wang et al., 2025) reports similar magnitudes. None of these works evaluate register preservation, since their target domains are register-homogeneous.

**Quality control and curriculum learning.** Quality control in LLM-based NER augmentation remains under-explored. (Kang et al., 2026) introduce a reflective multi-agent framework for tag consistency, but no prior framework jointly addresses structural, lexical, semantic, and *dialectal* quality. Curriculum learning (Bengio et al., 2009) has yielded gains in machine translation

(Platanios et al., 2019) and low-resource NLP (Xu et al., 2020), yet it has not been combined with LLM-augmentation quality scoring. Our work fills this gap.

### 3 Methodology

Our pipeline (Figure 2) has four stages: (I)  $k$ -shot seed sampling; (II) LLM-based sentence generation; (III) multi-dimensional quality characterisation; (IV) quality-aware curriculum training.

#### 3.1 $k$ -shot seed sampling

Sampling is restricted to the dialectal sub-corpus only: since Wojood 2024 is 82.9% MSA, global sampling would contaminate seed prompts with MSA-register sentences, propagating cross-register drift even before the generation step. We evaluate four shot sizes  $k \in \{5, 10, 15, 20\}$ , where  $k$  denotes the number of seed sentences drawn per entity-label type.

Two samplers are used. The GREEDY sampler stratifies by entity type with a rare-type priority boost and guarantees type coverage (Ye et al., 2024): sentences are grouped by entity type, up to  $k_{\max} = \lfloor 1.25k \rfloor$  examples are drawn per type, with  $k_{\max} \leftarrow 2k_{\max}$  for rare types (those representing less than 5% of the sub-corpus), and a final deduplication pass retains each sentence at most once across types. The SEMANTIC sampler enforces in-partition diversity via cosine-similarity filtering on distiluse-base-multilingual-cased embeddings (Reimers and Gurevych, 2019) with threshold  $\tau = 0.75$ , constructing a maximal independent set (Sener and Savarese, 2018). Both samplers run in  $O(|\mathcal{D}| + |\mathcal{Y}|)$  time, where  $|\mathcal{D}|$  is the corpus size and  $|\mathcal{Y}|$  the number of entity types. Full sampler details are in Appendix A.

#### 3.2 LLM-based augmentation

We use LLaMA-3.1-8B-Instruct (Grattafiori et al., 2024) with 4-bit NF4 quantisation (Dettmers et al., 2023) (BitsAndBytes, bfloat16). **Entity-level** augmentation replaces entity spans with new mentions of the same type while freezing surrounding context, preserving dialectal morphology by construction. **Context-level** augmentation rewrites the surrounding context using one of six Arabic stylistic directives while preserving entity spans verbatim, with diacritic-tolerant fuzzy verification. **Combined** augmentation applies the two sequentially, as illustrated in figure 1. All outputs pass a four-

stage filter (minimum length, language validity, entity match, deduplication) followed by BIO reconstruction; full details are in Appendix A.

#### 3.3 Multi-dimensional quality characterisation

$Q(s)$  is a per-sentence composite computed before training. We do not treat  $Q(s)$  as a generic synthetic-data filter; we use it as a diagnostic that decomposes generation quality into dimensions whose relative importance can be measured empirically and which, taken together, expose register drift that no single generic score captures.

**Configuration-level diagnostics** (dataset composition, lexical diversity, entity fidelity) inform strategy selection but do not enter  $Q(s)$ . **Seven per-sentence dimensions** form the score: three neural (LaBSE (Feng et al., 2022) semantic consistency, CAMeLBERT-Mix (Inoue et al., 2021) fluency, MARBERT (Abdul-Mageed et al., 2021) training utility); two lexical (novelty, repetition); a register-fidelity classifier (dialectal authenticity, DA); and a structural gate (IOB2 validity, EI). A supplementary **LLM-as-Judge** layer (Zheng et al., 2023) (Qwen2.5-7B-Instruct, five attributes on 1–5) is reported for cross-validation but excluded from  $Q(s)$  to avoid score conflation. Full definitions are in Appendix C. Each dimension is normalised to  $[0, 1]$  and combined as

$$Q(s) = \sum_i w_i d_i(s), \quad \sum_i w_i = 1. \quad (1)$$

Weights (Table 1) reflect failure-mode severity rather than ad-hoc tuning: DA and EI receive 0.20 each (register guard, structural gate); SC, FL, TU receive 0.15 each (neural semantics); LN and REP receive 0.10 and 0.05 (weaker proxies for NER utility). The score is conservative by construction: SC, FL and TU reward proximity to the seed distribution and carry 0.45 jointly, against 0.15 for the novelty dimensions. We substantiate the claim in §6.4 with a full one-dimension-out ablation, a uniform-weights baseline, and a threshold sweep.

#### 3.4 Quality curriculum training

Augmented sentences are sorted in descending  $Q(s)$  order and partitioned into three tiers: **Top-30%** (high-quality anchor), **Top-60%** (adds moderate-quality examples), and **All** (full set). Each configuration is trained four times, once per evaluation target (full Wojood 2024, Curras, Lebanese, both dialects), each run using the seeds

| Dim. | Tool                        | $w$  |
|------|-----------------------------|------|
| DA   | char TF-IDF + LR classifier | 0.20 |
| EI   | Binary IOB2 check           | 0.20 |
| SC   | LaBSE cosine similarity     | 0.15 |
| FL   | CAMeLBERT-Mix PLL           | 0.15 |
| TU   | MArBERT centroid proximity  | 0.15 |
| LN   | Novel content-token ratio   | 0.10 |
| REP  | Unique bigram ratio         | 0.05 |

Table 1: Per-sentence dimensions and weights in  $Q(s)$ .

of its own training set plus their augmentations. We additionally evaluate a **Unified** configuration pooling all three methods at  $Q(s) \geq 0.50$ . Every configuration is evaluated under two regimes: **flat**, which appends the augmented pool of that regime without tier ordering, and **curriculum**, which appends only the top-30% or top-60% tier. The contrast isolates quality ordering; per-configuration counts are in Table 10.

## 4 Data

All experiments use Wojood 2024 (Jarrar et al., 2024), with 21 entity types in flat BIO (Table 2). The dialectal content lies in Lebanese (298 train) and Curras (3,660 train), representing 17.1% of training data. Across all 24 configurations the pipeline yields **12,919 raw augmented sentences**, raising the dialectal share up to 23.6%. After filtering at  $Q(s) \geq 0.50$ , **10,674 sentences are retained** (82.6% overall). Curras retention is 94.7%, Lebanese 32.5%. The gap originates in Entity-Greedy generation on the small Lebanese seed pool, where substituted mentions break type compatibility with the gold span and score low on  $Q(s)$  (§6.7). Selection uses  $Q(s)$  alone; the LLM-as-Judge layer is diagnostic and never removes data.

| Sub-corpus Dialect         |                 | Train         | Val.         | Test         |
|----------------------------|-----------------|---------------|--------------|--------------|
| Lebanese                   | Lebanese Ar.    | 298           | 42           | 85           |
| Curras                     | Palestinian Ar. | 3,660         | 523          | 1,045        |
| Other (11 MSA sub-corpora) |                 | 19,167        | 2,739        | 5,476        |
| <b>Wojood total</b>        |                 | <b>23,125</b> | <b>3,304</b> | <b>6,606</b> |

Table 2: Wojood 2024 statistics.

## 5 Experimental Setup

**Configurations.** We cross method  $\in \{\text{Entity, Context, Both}\}$ , decoding  $\in \{\text{GREEDY, SEMANTIC}\}$ , and shot  $k \in \{5, 10, 15, 20\}$ , yielding 24 configurations on a single NVIDIA T4 GPU (16 GB VRAM).

**NER model.** We fine-tune ARBERTv2 (Abdul-Mageed et al., 2021), a 110M-parameter BERT for MSA and dialectal Arabic, on a 21-tag BIO scheme (AdamW,  $\text{lr} = 10^{-5}$ , 10 epochs). We report micro-F1 on entity spans for four training regimes, each with its own training set and its own test set: Full Wojood, Curras, Lebanese, and Dialectal (Curras+Lebanese).

**Baselines.** The **Wojood-only** baseline is Full WJ = 86.48, Curras = 70.35, Dialectal = 70.18, Lebanese = 0.00. The Lebanese zero is the score of a model trained on the 298 Lebanese seeds alone, which matches no gold span exactly: it is the floor the augmented Lebanese runs are compared against, not a failure of the full-corpus model. The **flat-augmentation** regime serves as a second baseline: it appends the augmented pool without tier ordering, so the comparison with curriculum isolates quality *ordering*; it is not an unfiltered condition.

**Additional controls.** (i) **Random top- $k\%$  selection:** a random subset of identical cardinality drawn from the quality-filtered pool ( $Q(s) \geq 0.50$ ), retrained identically, isolating ordering from volume reduction. (ii) **Second generator:** representative flat-augmentation configurations reproduced with Qwen2.5-7B-Instruct (same quantisation, prompts, filtering) to test backbone portability.

## 6 Results

### 6.1 Quality characterisation and the dialect preservation gap

All configurations achieve 100% IOB2 compliance after filtering, making structural correctness an engineering constraint rather than a differentiating signal. The central finding is the **dialect preservation gap** (Figure 4): entity-level augmentation scores  $\text{DA} = 0.88\text{--}0.94$ , as freezing the context preserves register by construction, while context-level and combined augmentation score  $\text{DA} = 0.45\text{--}0.59$ , near the 0.50 chance boundary, reflecting LLaMA’s strong MSA prior. Table 3 summarises per-method profiles. Entity augmentation leads on composite  $Q(s)$  and DA but scores only 1.70/5 on LLM-Judge due to single-token span compression (mean 1.23–1.50 tokens vs. 2.73 in Wojood 2024). Context augmentation receives the highest LLM-Judge score (3.42/5).

Figure 3 shows three representative sentence

| Method  | Sents        | Mean $Q(s)$ | Std  | $Q \geq 0.70$ | DA          | Judge       |
|---------|--------------|-------------|------|---------------|-------------|-------------|
| Entity  | <b>6,771</b> | <b>0.72</b> | 0.04 | <b>75.2%</b>  | <b>0.92</b> | 1.70        |
| Context | 3,436        | 0.65        | 0.08 | 30.4%         | 0.51        | <b>3.42</b> |
| Both    | 467          | 0.63        | 0.08 | 22.9%         | 0.53        | 2.58        |

Table 3: Per-method quality summary. **Bold**: best per column.

| Original (dialect)                                                                                                       | Augmented                                                                                                                                                  | Issue                                                                              |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| الدكتور : واضح انه صحته كويسة<br>(Doctor: it is clear that his health is good)<br><i>Lebanese; informal wording</i>      | كان الباحث واضحاً في أن كان مصاباً<br>بصدمة كبيرة<br>(The researcher was clear that he was suffering from a major shock)<br><i>MSA lexical/style shift</i> | <b>Register drift</b><br>(MSA contamination)<br>Register shifts toward formal MSA. |
| فريق الريال<br>(Real team)<br><i>2-token org</i>                                                                         | برشلونة<br>(Barcelona)<br><i>1-token org</i>                                                                                                               | <b>Span compression</b><br>Named entity span shortened.                            |
| الموظف : حسب الكتيب الفنان 10%<br>(Employee: for artists, the rate in the handbook is 10%)<br><i>3 entities retained</i> | رأيت الفنان على المسرح، فوجئت بمواهه في التمثيل<br>(I saw the artist on stage, and I was surprised by his acting talent)<br><i>1 entity retained</i>       | <b>Entity loss</b><br>Two entities dropped ("al-kateeb" and "10%").                |

Figure 3: Representative augmentation failures. Full examples in Appendix H.

pairs that make the gap concrete. The Lebanese example is particularly diagnostic: every entity is preserved, fluency is high, and LaBSE similarity remains above 0.85, so a generic synthetic-data filter would accept the rewrite, yet the dialect itself has been replaced by newswire MSA, a case of augmentation contamination that fluency and similarity scores cannot detect. Section 6.7 returns to these failure modes quantitatively.

Two caveats qualify the gap. The entity-level score is near-tautological, as such a sentence differs from its seed only in its mentions. The context-level score, conversely, bounds spontaneous drift from above rather than measuring it: the rewrite prompts require Arabic output but never the seed register, and one of the six sampled styles explicitly requests a formal journalistic one (Appendix B).

## 6.2 NER results: flat augmentation

Table 4 reports the strongest configurations per method; the full 24-configuration grid is in Appendix E.

Three findings stand out. (i) *Entity augmentation dominates Lebanese, with orthogonal sampling strategies for Curras vs. Lebanese*. Entity-greedy at 10 and 20 shots reaches Lebanese F1 = 50.00, the highest across all regimes, while entity-semantic-10 achieves the best Curras peak (76.93, +6.58 pp). With 85 test sentences this F1 orders regimes rather than calibrating them. It is nonetheless where the synthetic sentences show most clearly, no MSA data entering it: the 298

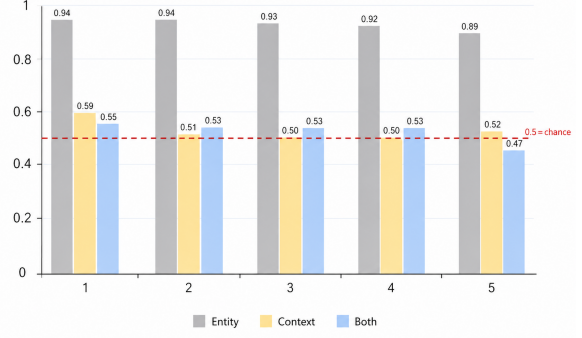


Figure 4: Dialect preservation gap across augmentation methods. Entity augmentation preserves register by design; context and combined augmentation drift toward MSA.

seeds alone give 0.00, the seeds plus their entity-greedy augmentations 40.98, 50.00 and 50.00 at 5, 10 and 20 shots. (ii) *Context augmentation is the only method to improve Full Wajood F1* (87.02–87.18, +0.54 to +0.70 pp). (iii) *Both augmentation has the narrowest profile*, with low volumes (21–106 sentences per configuration) and zero Lebanese F1, showing that the sequential constraint limits rather than compounds the individual strategies.

## 6.3 NER results: quality curriculum training

Three patterns emerge. First, quality filtering stabilises Full Wajood F1, with per-configuration curriculum peaking at context-greedy-15 top30 (87.45, +0.97 pp). **Context-top60** offers the best practical trade-off: Curras F1 = 76.05 (+5.70 pp), Dialectal F1 = 74.85 (+4.67 pp), and Lebanese F1 = 36.97 at only −0.71 pp on Full Wajood. Unified-top60 reaches a higher Lebanese F1 (43.48) but concedes 1.72 pp on Curras and 0.99 pp on Full Wajood; Context-top60 is the only tier that improves every dialectal column at once. The non-monotonic tier effect (Figure 5; top60 > top30 > all for entity, top60 > all > top30 for context) supports the curriculum hypothesis: quality matters more than quantity, and both overly strict and overly broad inclusion degrade performance; §6.4 discusses a competing reading of the same profile. Entity-Lebanese collapses to zero under the top tiers: the top tiers retain almost none of them, and the span-compression bias of §6.7 dominates what survives. These observations are consistent with the statistical analysis in Appendix F, which confirms robust Curras gains across all configurations, while Lebanese results are highly variable and in-

| Config             | Aug.  | Full Wojood  |              | Curras       |              | Dialectal    |              | Lebanese <sup>†</sup> |              |
|--------------------|-------|--------------|--------------|--------------|--------------|--------------|--------------|-----------------------|--------------|
|                    |       | F1           | $\Delta$     | F1           | $\Delta$     | F1           | $\Delta$     | F1                    | $\Delta$     |
| Baseline           | —     | 86.48        | —            | 70.35        | —            | 70.18        | —            | 0.00                  | —            |
| entity-greedy-10   | 1,007 | 85.76        | −0.72        | 75.56        | +5.21        | 74.18        | +4.00        | <b>50.00</b>          | <b>50.00</b> |
| entity-greedy-20   | 1,589 | 85.95        | −0.53        | 75.03        | +4.68        | 73.87        | +3.69        | <b>50.00</b>          | <b>50.00</b> |
| entity-semantic-10 | 522   | 86.05        | −0.43        | <b>76.93</b> | <b>+6.58</b> | <b>74.52</b> | <b>+4.34</b> | 6.90                  | 6.90         |
| context-greedy-5   | 310   | <b>87.18</b> | <b>+0.70</b> | 75.38        | +5.03        | 73.26        | +3.08        | 16.67                 | 16.67        |
| context-greedy-10  | 547   | 87.03        | +0.55        | 76.34        | +5.99        | 73.24        | +3.06        | 36.04                 | 36.04        |
| context-greedy-15  | 713   | 87.14        | +0.66        | 76.88        | +6.53        | 73.72        | +3.54        | 19.05                 | 19.05        |
| both-greedy-5      | 34    | 85.91        | −0.57        | 74.26        | +3.91        | 73.36        | +3.18        | 0.00                  | —            |
| both-semantic-10   | 35    | 85.93        | −0.55        | 74.18        | +3.83        | 73.64        | +3.46        | 0.00                  | —            |

Table 4: Flat augmentation: selected configurations per method.  $\Delta$  vs. baseline (86.48 / 70.35 / 70.18 / 0.00). <sup>†</sup>: Lebanese baseline is zero so  $\Delta$  equals F1. **Bold**: column-best.

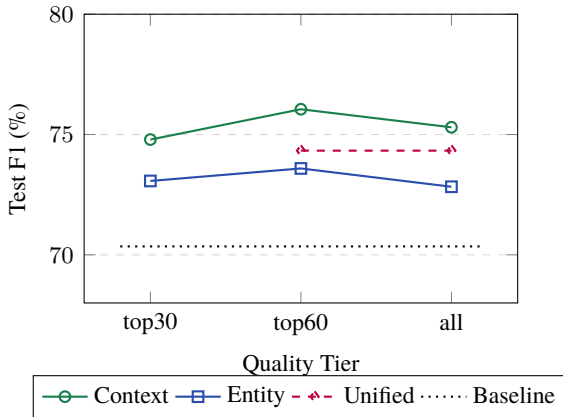


Figure 5: Curras F1 across quality tiers. Both methods peak at top60 (entity: top60 > top30 > all; context: top60 > all > top30), and the non-monotonic profile supports the curriculum hypothesis.

clude zero outcomes that reflect instability rather than a true absence of improvement.

#### 6.4 $Q(s)$ ablation and robustness

We verify that the choice of  $Q(s)$  dimensions and weights is not arbitrary, and that the framework is more than a re-labelling of generic synthetic-data filtering, through three families of follow-up experiments on the Entity-method pool, sharing baselines with Table 12 in Appendix E. (i) We ablate one dimension at a time, redistributing its weight uniformly. (ii) We replace the proposed weighting with Uniform 1/7 and a DA-heavy variant. (iii) We sweep the threshold to  $Q \geq 0.60$  and  $Q \geq 0.70$ . Table 5 reports the headline results; the complete grid, Spearman ranking matrix, Jaccard top-tier overlaps, and per-subcorpus  $Q(s)$  means are in Appendix G.

| Variant         | full wojood  |       | Curras       |       | Lebanese     |       |
|-----------------|--------------|-------|--------------|-------|--------------|-------|
|                 | $Q \geq .50$ | top30 | $Q \geq .50$ | top30 | $Q \geq .50$ | top30 |
| Original $Q(s)$ | 87.24        | 84.87 | 74.39        | 73.07 | <b>28.07</b> | 0.00  |
| −DA             | 87.12        | 87.20 | 75.40        | 75.08 | 7.06         | 4.71  |
| −EI             | 87.26        | 87.10 | 74.35        | 75.22 | 28.07        | 2.44  |
| −FL             | 87.29        | 87.12 | 74.05        | 75.48 | 25.69        | 2.41  |
| −TU             | 87.11        | 87.32 | 73.55        | 75.01 | 19.64        | 11.24 |
| Uniform 1/7     | 87.16        | 87.22 | 74.86        | 74.80 | 4.76         | 2.44  |
| DA-heavy 0.40   | 87.10        | 87.20 | 74.04        | 74.86 | 26.32        | 0.00  |

Table 5:  $Q(s)$  ablation on the Entity-method pool. Both regimes share their baseline with Table 12.

**Each dimension contributes asymmetrically.** At the operational  $Q \geq 0.50$  threshold, removing **DA** produces the single largest Lebanese F1 collapse (28.07 to 7.06, a 21 pp drop): without a register-fidelity signal the filter silently re-admits MSA-contaminated sentences, and the only subcorpus depending on register fidelity loses its augmented support. Removing **TU** produces a smaller Lebanese drop (−8 pp) and the largest ranking disruption (Spearman = 0.86), confirming that MARBERT-centroid proximity captures a signal no other dimension reproduces. Removing **FL** affects mainly the high-quality tail (−2.4 pp on Lebanese). Removing **EI** does not noticeably change the results at this threshold since post-filter sentences already satisfy IOB2. No single ablation reproduces the original Lebanese F1, while Uniform weighting pushes it below 5. Full Wojood and Curras stay within  $\pm 0.5$  pp of baseline across all variants: the weighting acts selectively on the most fragile subcorpus. This asymmetry confirms that register fidelity is not recoverable from fluency or similarity alone: a filter relying solely on those signals would silently admit MSA-contaminated sentences, replicating the −DA collapse rather than preventing it.

| Config            | Metric      | Tier  | Flat  | Q-score      | Random | $\Delta_{QR}$ |
|-------------------|-------------|-------|-------|--------------|--------|---------------|
| entity-greedy-15  | Full Wojood | top30 | 85.93 | 87.02        | 86.61  | +0.41         |
| entity-greedy-15  | Dialectal   | top60 | 73.78 | 72.73        | 72.82  | -0.09         |
| entity-semantic-5 | Curras      | top30 | 71.32 | 74.50        | 73.63  | +0.87         |
| both-semantic-15  | Curras      | top30 | 74.47 | <b>75.62</b> | 73.59  | <b>+2.03</b>  |
| both-greedy-20    | Full Wojood | top60 | 86.00 | 87.28        | 86.65  | +0.63         |
| context-greedy-10 | Lebanese    | top30 | 36.04 | <b>20.41</b> | 16.84  | <b>+3.57</b>  |
|                   |             | top60 |       | <b>24.53</b> | 0.00   | <b>+24.53</b> |

Table 6: Q-score curriculum vs. random top- $k\%$  selection.  $\Delta_{QR}$  = Q-score – Random. **Bold**:  $\geq 1$  pp gap.

**Robustness.** Uniform  $1/7$  weights crash Lebanese by 23 pp while leaving Curras and Full Wojood intact, confirming that the loss concentrates where dialectal signal matters most. A DA-heavy variant ( $w_{DA} = 0.40$ ) is essentially equivalent to the original ( $-1.8$  pp on Lebanese), so the information is in DA being weighted, not in its precise magnitude. Threshold sensitivity is sharper: tightening to  $Q \geq 0.60$  or  $Q \geq 0.70$  collapses Lebanese F1 to zero (Appendix G), fixing  $Q \geq 0.50$  as the operating point that preserves enough low-resource content for downstream gains. One caveat follows from the weighting: with 0.45 on similarity against 0.15 on novelty, the top tiers favour conservative rewrites. Removing TU costs 8 pp on Lebanese, so moderate conservatism helps, but the non-monotonic tier profile is equally consistent with the top-30% tier sitting too close to its seeds; a per-tier novelty mean would separate the two readings.

## 6.5 Q-score vs. random-selection control

To isolate  $Q(s)$ ’s contribution from mere data-volume reduction, we compare each  $Q(s)$ -ordered top- $k\%$  tier against a randomly drawn subset of identical cardinality from the same filtered pool (Table 6).

$Q(s)$ ’s advantage over random selection scales with sub-corpus fragility. On Curras and Full Wojood the advantage is positive but moderate. On Lebanese, a random subset collapses F1 entirely at the top-60% tier of context-greedy-10 while the  $Q(s)$ -ordered tier retains a substantial curriculum gain. The mechanism is structural: random selection inherits the bottom  $Q(s)$  tail proportionally, and that tail concentrates the entity-faithfulness violations of §6.7, which hurt Lebanese disproportionately while resource-richer corpora dilute their

| Config             | Generator | Full Wojood  | Curras       | Lebanese     | Dialectal    |
|--------------------|-----------|--------------|--------------|--------------|--------------|
| entity-semantic-10 | LLaMA     | 86.05        | <b>76.93</b> | 6.90         | <b>74.52</b> |
|                    | Qwen      | <b>87.22</b> | 74.53        | 0.00         | 73.46        |
| entity-greedy-10   | LLaMA     | 85.76        | <b>75.56</b> | <b>50.00</b> | 74.18        |
|                    | Qwen      | <b>87.20</b> | 75.39        | 24.30        | <b>74.38</b> |
| context-greedy-5   | LLaMA     | 87.18        | <b>75.38</b> | 16.67        | 73.26        |
|                    | Qwen      | <b>87.27</b> | 75.27        | <b>24.30</b> | <b>74.32</b> |
| Baseline           | —         | 86.48        | 70.35        | 0.00         | 70.18        |

Table 7: Flat-augmentation results for LLaMA-3.1-8B vs. Qwen2.5-7B-Instruct.

effect.  $Q(s)$  therefore acts as a selective register-and-fidelity guard whose protective function scales with sub-corpus fragility.

## 6.6 Generator generalisation

To test whether pipeline gains depend on LLaMA-3.1-8B-Instruct, we reproduce three flat-augmentation configurations, spanning both strategies and sampling paradigms, with Qwen2.5-7B-Instruct (identical quantisation, prompts, and filtering; Table 7). The configurations cover LLaMA-3.1-8B-Instruct’s strongest results on Full Wojood (*context-greedy-5*), Curras (*entity-semantic-10*), and Lebanese (*entity-greedy-10*).

Three observations follow. (i) *The pipeline is generator-portable on resource-sufficient sub-corpora.* Qwen meets or exceeds LLaMA-3.1-8B-Instruct on Full Wojood across all three configurations, and both backbones improve Curras by five to seven points above baseline. (ii) *The Lebanese ranking is strategy-dependent.* Under entity augmentation, LLaMA-3.1-8B-Instruct leads decisively (entity-greedy-10: 50.00 vs. 24.30), because entity substitution requires producing Lebanese-specific surface forms sensitive to the generator’s dialectal lexical inventory. Under context augmentation, where entity spans are preserved verbatim, Qwen matches or exceeds LLaMA-3.1-8B-Instruct (context-greedy-5: 24.30 vs. 16.67). The discriminating factor is task specificity, not aggregate generator quality. (iii) *A generator-dependent register ceiling emerges.* This generator–strategy interaction is unique to Lebanese and becomes legible only under  $Q(s)$ -driven sub-corpus decomposition, making generator selection a register-preservation design variable for low-resource dialectal augmentation.

## 6.7 Augmentation failure modes

The errors below are properties of the *augmented data*, not predictions of the trained NER model. Representative examples are in Figure 3 and Appendix H.

**Span compression (entity method).** Generated entity spans are markedly shorter than the corpus average: 60–81% are single-token, and mean span length falls to 1.23–1.50 tokens against 2.73 in Wojood 2024. This generation-side bias propagates to the trained model as a single-token boundary preference, explaining the collapse of Lebanese entity-curriculum F1 from 50.00 (flat) to zero (top30/top60).

**Cross-register drift (context method).** Context rewriting yields  $DA \approx 0.45\text{--}0.59$ : roughly half of generated sentences are classified as MSA. Dialectal function-word substitution and MSA-order rewriting occur while entity spans are preserved correctly. These sentences still improve Curras under flat augmentation because contextual diversity outweighs the register cost; they do not generalise to Lebanese, whose tighter register distribution makes  $Q(s)$ 's DA component the deciding filter at the high-quality tier.

**Entity-faithfulness violations.** Entity-Greedy retains only 10.8% of Lebanese augmented sentences vs. 100% on Curras. With 298 Lebanese seeds, substituted entities frequently violate type semantics (e.g., a village name replaced by a country name, both gold-type GPE but at incompatible granularities).  $Q(s)$  filters this failure mode at the operational threshold and explains the random-baseline collapse in §6.5: a random 60% subset retains these violations while  $Q(s)$  top-60% discards them.

## 7 Discussion

**Register fidelity vs. contextual diversity.** Entity augmentation preserves register ( $DA = 0.88\text{--}0.94$ ), yet context augmentation produces more consistent downstream gains despite MSA drift, by placing entities in more varied syntactic frames and yielding more than three times as many Lebanese sentences that survive filtering (586 vs. 175 across the eight configurations of each method, Table 8). Contextual diversity is therefore a more effective signal than strict register fidelity *once drift is bounded*: at  $DA \approx 0.5$  the gain holds on

Curras but not on the tighter Lebanese register, where  $Q(s)$ 's DA component becomes the deciding filter. Span compression further explains entity augmentation's curriculum failure on Lebanese.

**Flat augmentation vs. curriculum.** The two regimes optimise complementary objectives. Flat augmentation delivers the highest peaks (Curras 76.93; Lebanese 50.00); curriculum trades peaks for consistency, reducing entity Curras F1 standard deviation from 1.86 to 0.62 (top30) while every per-configuration curriculum run stays within 86.4–87.5 on Full Wojood (Table 13).

Three properties make  $Q(s)$  a register-aware diagnostic rather than a generic filter: it is *domain-specific* (DA classifier has no analogue in standard NER filters), *selective* (removing any single dimension leaves easier sub-corpora within  $\pm 0.5$  pp while shifting Lebanese by up to 23 pp), and *empirically necessary* (a fluency- or similarity-only filter keeps Curras gains but loses Lebanese). The random-selection and generator-comparison controls further confirm that quality ordering and generator selection address complementary failure modes within the same diagnostic frame.

## 8 Conclusion

We presented a register-aware LLM augmentation pipeline for Arabic dialectal NER. Our composite score  $Q(s)$  quantifies a dialect preservation gap between entity- and context-level rewriting and orders the augmented pool for curriculum training. Flat entity augmentation produces the Lebanese peak (0 to 50.00); curriculum ordering trades it for stability (+5.70 pp on Palestinian, 36.97 on Lebanese). Component ablation, weight-sensitivity, threshold sweep, and a random-selection control indicate that the register-fidelity dimension carries most of  $Q(s)$ 's protective effect, and that quality ordering, not volume, accounts for the curriculum gains. A second LLM backbone reproduces these gains while exposing a generator-dependent register ceiling, framing generator selection itself as a register-preservation design variable.

## Limitations

**Evaluation on the Lebanese sub-corpus.** The Lebanese sub-corpus provides only 298 training and 85 test sentences. At that size micro-F1 moves in large steps: 50.00 recurs across configurations

while neighbours land on 16.67 or 6.90, so the Lebanese column orders regimes rather than calibrating them. [Berg-Kirkpatrick et al. \(2012\)](#) show that evaluation is less reliable on small test sets, and [Dror et al. \(2018\)](#) recommend bootstrap testing there. Ours resamples configurations, not test sentences, so it describes the spread of the flat-augmentation grid rather than the uncertainty of one system: 9 of the 23 configurations exceed the zero baseline and 14 do not, giving a configuration-level mean of 11.88 against 47.44 for entity-greedy alone. Each configuration ran under a single seed, leaving training variance unmeasured, and configurations are compared directly on the test sets, so the per-method peaks are not selection-free.

**Training-side ceiling.** No configuration generates more than 443 Lebanese sentences or keeps more than 220 after filtering, and in the largest both-level configuration only 4 of 106 retained augmentations are Lebanese. Data at this volume can shift the training distribution but cannot replace real Lebanese sentences. On the full corpus they sit beside 19,167 MSA ones, so their contribution is not separable from the Full Wojood column.

**Scope of the register measurement.** DA is trained on sub-corpus provenance, so it measures similarity to dialectal Wojood text as written rather than dialect identity, and we report no evaluation on independently register-labelled data. The prompts require Arabic output but not the seed register, and one of the six rewrite styles asks for a formal journalistic register; the DA  $\approx 0.51$  of context-level rewriting therefore bounds spontaneous drift from above rather than isolating it.

**Generation and decoding.** LLaMA-3.1-8B-Instruct shows a strong MSA prior in context-level generation under these prompts, and entity augmentation produces compressed spans (1.23–1.50 tokens) that do not generalise to the multi-word entities of the Lebanese test set.  $Q(s)$  is applied post hoc: generation is never optimised against it. Four directions follow: dialect-specific  $Q(s)$  thresholds, constrained decoding for multi-token spans, best-of- $n$  sampling under  $Q(s)$ , and generator selection on target-dialect lexical fidelity; extension to Egyptian, Algerian and other Maghrebi dialects would further test generalisability.

## Ethics Statement

This work uses publicly available datasets (Wojood 2024, LaBSE, CAMeLBERT, MARBERT) under their respective licences. Augmented data is generated from existing annotated text and introduces no new personally identifiable information. Compute was performed on publicly available academic GPU resources.

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## A Post-Generation Filtering Pipeline and Samplers

All generated outputs undergo a two-stage processing pipeline before being included in the augmented dataset. First, a four-stage filter removes invalid or redundant sentences and reconstructs their BIO annotations. Second, a sampler selects a diverse and balanced subset from the retained outputs for final export.

### A.1 Filtering Pipeline

**Four-stage filter.** Each generated sentence is passed through four successive filters:

1. **Length.** Sentences shorter than 10 characters are discarded.

2. **Language validity.** Sentences containing Chinese characters or with an Arabic character ratio below 30% are rejected.

3. **Entity match.** For entity-level outputs, at least one replaced entity must correspond to a known ground-truth entity; for context-level outputs, all original entities must appear in the rewritten sentence under diacritic-tolerant match.

4. **Deduplication.** Augmented sentences identical to the original or to a previously accepted variant are discarded.

**BIO reconstruction.** Sentences that pass all four filters are reconstructed into BIO sequences via a three-step cascade: exact match, punctuation-stripped match, and diacritic-insensitive fuzzy match. A final validator corrects any residual BIO inconsistencies in the augmented sentences before they are forwarded to the sampling stage.

### A.2 Samplers

Once filtering and BIO reconstruction are complete, a sampler selects the final subset of augmented examples according to one of two strategies: a greedy entity-balanced sampler (GREEDY) or a semantic diversity sampler (SEMANTIC).

**Greedy sampler (GREEDY).** GREEDY groups sentences by entity type and draws up to  $k_{\max} = \lfloor 1.25k \rfloor$  examples per type. Rare entity types (those representing less than 5% of sentences) receive a priority boost of  $2k_{\max}$  to prevent underrepresentation. A final deduplication pass ensures that each sentence appears at most once in the selected subset.

**Semantic diversity sampler (SEMANTIC).** SEMANTIC promotes lexical and semantic variety in the selected examples. Each candidate sentence  $s$  is encoded with distiluse-base-multilingual-cased, and is admitted into the selected subset  $\mathcal{D}_{\text{smp}}$  only if its cosine similarity to all previously selected sentences falls below a threshold  $\tau = 0.75$ :

$$\max_{s' \in \mathcal{D}_{\text{smp}}} \cos(\mathbf{e}_s, \mathbf{e}_{s'}) < \tau.$$

This ensures that semantically redundant augmentations are excluded from the final dataset.

## B Augmentation Prompt Templates

Figures 6–8 show the prompt templates used for the entity-level, context-level, and both-level augmentation strategies. Figure 9 provides the entity-type definitions injected into each prompt.

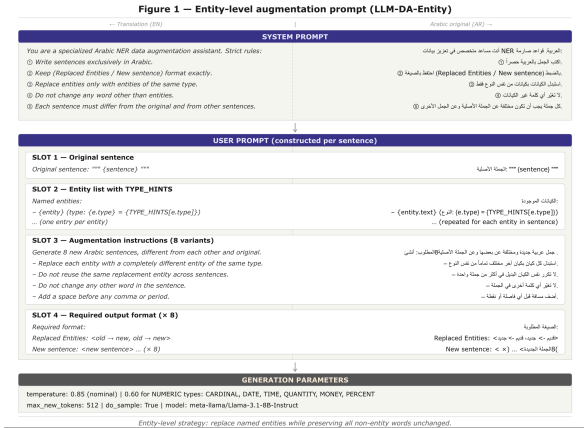


Figure 6: Entity-level augmentation prompt (LLM-DA-Entity).

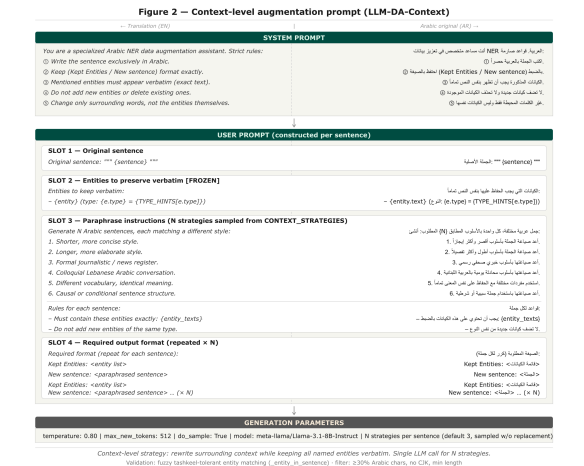


Figure 7: Context-level augmentation prompt (LLM-DA-Context).

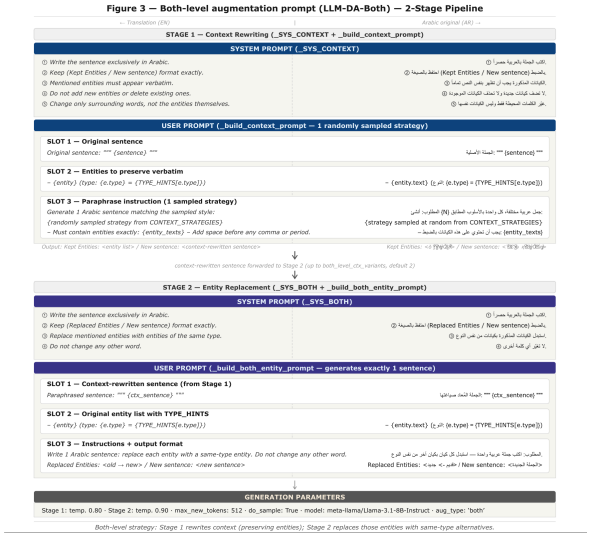


Figure 8: Both-level augmentation prompt (LLM-DA-Both).

Table – TYPE\_HINTS: NER entity type definitions (Arabic + English)

| Type        | Arabic definition (TYPE_HINTS)              | English translation                        |
|-------------|---------------------------------------------|--------------------------------------------|
| GPE         | اسم بلد أو مدينة أو منطقة جغرافية           | Country, city, or geopolitical area        |
| LOC         | اسم موقع أو مكان جغرافي                     | Location or geographical place             |
| ORG         | اسم منظمة أو مؤسسة أو شركة                  | Organization, institution, or company      |
| PER         | اسم شخص                                     | Person name                                |
| NORP        | اسم جنسية أو مجموعة إثنية أو سياسية         | Nationality, religious, or political group |
| CARDINAL    | رقم أو عدد (مثلاً: خمسة، اثنا عشر، ألف)     | Cardinal number (e.g. five, 12, thousand)  |
| DATE        | تاريخ أو سنة أو فترة زمنية                  | Date, year, or time period                 |
| TIME        | وقت أو ساعة                                 | Time of day or hour                        |
| QUANTITY    | كمية مع وحدة قياس (مثلاً: ثلاثون كيلومتراً) | Quantity with unit (e.g. 30 kilometres)    |
| MONEY       | مبلغ مالي مع عملة (مثلاً: خمسمئة دولار)     | Monetary amount with currency (e.g. \$500) |
| PERCENT     | نسبة مئوية (مثلاً: ثلاثون بالمئة)           | Percentage (e.g. 30%)                      |
| EVENT       | اسم حدث أو مناسبة                           | Named event or occasion                    |
| LANGUAGE    | اسم لغة                                     | Language name                              |
| PRODUCT     | اسم منتج                                    | Product name                               |
| LAW         | اسم قانون أو اتفاقية                        | Law, regulation, or agreement              |
| WORK_OF_ART | اسم عمل فني أو أدبي                         | Artistic or literary work                  |

Named entity types (GPE, LOC, ORG, PER, NORP, EVENT, LANGUAGE, PRODUCT, LAW, WORK\_OF\_ART)  
Numeric types – lower generation temperature (0.60) to reduce hallucination

Figure 9: TYPE\_HINTS: NER entity-type definitions.

## C Quality Dimension Definitions

Table 20 gives full definitions for each configuration-level diagnostic and per-sentence metric that constitutes the composite score  $Q(s)$ . One caveat applies to the DA dimension. Its supervision comes from sub-corpus provenance: positives are the Lebanese and Curras sentences of Wojood, negatives the MSA sub-corpora. DA therefore scores similarity to dialectal Wojood text as written, and we report no evaluation on independently register-labelled data. It is used as a comparative signal between augmentation methods and between quality tiers, not as a calibrated dialect identifier.

## D Detailed Augmentation Statistics

Table 8 summarises raw vs. quality-filtered counts by method and strategy. Tables 9 and 10 give the full 24-configuration training-set statistics before and after  $Q(s) \geq 0.50$  filtering. Counts in both tables are training-set totals for the sub-corpus, that is the base sentences plus the augmented ones retained at that stage; the number of augmented sentences is the difference from the baseline row.

## E Full Per-Configuration NER Results

This appendix presents the complete per-configuration experimental results across all training regimes and test sets. Table 11 reports the micro-F1 scores and performance changes ( $\Delta$ ) for all 24 individual configurations under the flat augmentation regime. Table 12 summarises the performance of quality curriculum training categorised by augmentation method and quality tier. Table 13 details the micro-F1 scores for each of the 24 configurations trained on the top-30% and top-60% tiers.

## F Statistical Reliability of Augmentation Gains

We apply bootstrap resampling (10,000 iterations) and binomial sign tests across the 23 flat-augmentation configurations other than context-greedy-20. The resampling unit is the configuration, not the test sentence: the intervals below describe the dispersion of the configuration grid and are not confidence intervals for the score of a single trained system. Sentence-level uncertainty on the 85-sentence Lebanese test set, and seed-level variance, are not estimated here. Curras gains are robust: bootstrap mean F1 = 74.60 (95% CI [74.06, 75.12]), all 23 configurations exceed baseline (sign test  $p < 0.0001$ ). Lebanese is bimodal: 9 of the 23 configurations exceed the zero baseline and 14 produce zero F1, giving a configuration-level mean of 11.88 against 47.44 for entity-greedy alone; the sign test over configurations is correspondingly non-significant. Entity-greedy significantly outperforms context-greedy on Lebanese (means 47.44 vs. 23.92,  $p < 0.0001$ ). On Curras, entity-semantic and entity-greedy are statistically indistinguishable ( $p = 0.72$ ). Curriculum training reduces entity-Curras standard deviation from 1.86 to 0.62 (top-30%).

## G Full $Q(s)$ Ablation and Sensitivity

All ablation experiments use the Entity-method pool. The original-weights baseline at top-30% is the “Entity top30” row of Table 12 (84.87 / 73.07 / 0.00); at  $Q \geq 0.50$  it is the threshold-filter run on the same pool (87.24 / 74.39 / 28.07). Variants: (i) **dimension ablations** (−DA, −EI, −FL, −TU); (ii) **weight perturbations** ( $\pm 10\%$  each, plus DA-heavy 0.40 and Uniform 1/7); (iii) **threshold sweep** at  $Q \geq 0.50, 0.60, 0.70$ .

Table 14 reports the full NER ablation grid (dimension ablations and weight variants) together with the threshold sensitivity in its bottom panel. Table 15 reports the Spearman rank correlation between all  $Q(s)$  variants; Tables 16 and 17 report Jaccard top-tier overlaps and per-subcorpus mean  $Q(s)$ , respectively.

## H Representative Augmentation Errors

To complement the aggregate metrics and illustrate the failure modes described in §6.7, we manually inspected the 1,250 entity-level Curras outputs (entity-greedy-15) and the 106 both-level outputs (both-greedy-20) used in our experiments. We detail the four recurring qualitative failure modes below.

**Span compression on dialectal compositional entities.** Although the aggregate fraction of multi-token entities among the matched Curras spans is largely preserved (27.0% in the Curras seed vs. 32.8% after entity-level augmentation), a specific dialectal construction is reliably collapsed: periphrastic entities of the form *head + proper noun* or *kin/group-term + place* are rewritten as a single proper noun of the same broad type. For example, the 2-token ORG span “فريق الريال” (lit. “*the Real team*”) is replaced across 14 augmentations by single-token club names (بنفيكا, برشلونة, يوفنتوس, ...), and the 2-token NORP span “اهل الضفة” (“*people of the West Bank*”) becomes single-token demonyms (السعوديين, الفلسطينيين, السودان). Among the 185 multi-token Curras entities for which a parent context could be matched, 9 (4.9%) were replaced by single-token spans of the same type.

### Register drift toward Modern Standard Arabic.

Both- and context-level rewrites systematically substitute Levantine morphological cues with MSA constructions, even though the rewrite prompts require Arabic output without requiring the seed register. The clearest case is Lebanese

| Method  | Strategy | Raw    |        |       | Filtered ( $\geq 0.50$ ) |        |      | Retention |       |
|---------|----------|--------|--------|-------|--------------------------|--------|------|-----------|-------|
|         |          | Total  | Curras | Leb.  | Total                    | Curras | Leb. | Curras    | Leb.  |
| Entity  | Greedy   | 5,752  | 4,354  | 1,398 | 4,505                    | 4,354  | 151  | 100.0%    | 10.8% |
| Entity  | Semantic | 2,369  | 2,242  | 127   | 2,266                    | 2,242  | 24   | 100.0%    | 18.9% |
| Context | Greedy   | 3,111  | 2,389  | 722   | 2,444                    | 1,893  | 551  | 79.2%     | 76.3% |
| Context | Semantic | 1,147  | 992    | 155   | 992                      | 957    | 35   | 96.5%     | 22.6% |
| Both    | Greedy   | 386    | 293    | 93    | 324                      | 273    | 51   | 93.2%     | 54.8% |
| Both    | Semantic | 154    | 139    | 15    | 143                      | 139    | 4    | 100.0%    | 26.7% |
| All     |          | 12,919 | 10,409 | 2,510 | 10,674                   | 9,858  | 816  | 94.7%     | 32.5% |

Table 8: Raw vs. quality-filtered augmented sentence counts.

| Method   | Strategy | Shots | Total  | Aug.  | Dial.% | $\Delta$ dial. | Curras | Lebanese |
|----------|----------|-------|--------|-------|--------|----------------|--------|----------|
| Entity   | Greedy   | 5     | 23,949 | 824   | 20.0%  | +2.9           | 4,254  | 528      |
|          |          | 10    | 24,472 | 1,347 | 21.7%  | +4.6           | 4,643  | 662      |
|          |          | 15    | 24,736 | 1,611 | 22.5%  | +5.4           | 4,910  | 659      |
|          |          | 20    | 25,095 | 1,970 | 23.6%  | +6.5           | 5,187  | 741      |
|          | Semantic | 5     | 23,417 | 292   | 18.1%  | +1.0           | 3,941  | 309      |
|          |          | 10    | 23,680 | 555   | 19.1%  | +2.0           | 4,172  | 341      |
|          |          | 15    | 23,844 | 719   | 19.6%  | +2.5           | 4,349  | 328      |
|          |          | 20    | 23,928 | 803   | 19.9%  | +2.8           | 4,420  | 341      |
| Context  | Greedy   | 5     | 23,534 | 409   | 18.6%  | +1.5           | 3,959  | 408      |
|          |          | 10    | 23,881 | 756   | 19.7%  | +2.6           | 4,196  | 518      |
|          |          | 15    | 24,026 | 901   | 20.2%  | +3.1           | 4,352  | 507      |
|          |          | 20    | 24,170 | 1,045 | 20.7%  | +3.6           | 4,522  | 481      |
|          | Semantic | 5     | 23,386 | 261   | 18.0%  | +0.9           | 3,801  | 418      |
|          |          | 10    | 23,366 | 241   | 18.0%  | +0.9           | 3,891  | 308      |
|          |          | 15    | 23,408 | 283   | 18.1%  | +1.0           | 3,933  | 308      |
|          |          | 20    | 23,487 | 362   | 18.4%  | +1.3           | 4,007  | 313      |
| Both     | Greedy   | 5     | 23,175 | 50    | 17.3%  | +0.2           | 3,688  | 320      |
|          |          | 10    | 23,210 | 85    | 17.4%  | +0.3           | 3,727  | 316      |
|          |          | 15    | 23,244 | 119   | 17.5%  | +0.4           | 3,756  | 321      |
|          |          | 20    | 23,257 | 132   | 17.6%  | +0.5           | 3,762  | 328      |
|          | Semantic | 5     | 23,149 | 24    | 17.2%  | +0.1           | 3,681  | 301      |
|          |          | 10    | 23,164 | 39    | 17.3%  | +0.2           | 3,694  | 303      |
|          |          | 15    | 23,163 | 38    | 17.3%  | +0.2           | 3,695  | 301      |
|          |          | 20    | 23,178 | 53    | 17.3%  | +0.2           | 3,709  | 302      |
| Baseline |          |       | 23,125 | –     | 17.1%  | –              | 3,660  | 298      |

Table 9: Full raw augmented training-set statistics (before filtering).

| Method   | Strategy | Shots | Total  | Aug.  | Dial.% | $\Delta$ dial. | Curras | Lebanese |
|----------|----------|-------|--------|-------|--------|----------------|--------|----------|
| Entity   | Greedy   | 5     | 23,740 | 615   | 19.3%  | +2.2           | 4,254  | 319      |
|          |          | 10    | 24,132 | 1,007 | 20.6%  | +3.5           | 4,643  | 322      |
|          |          | 15    | 24,419 | 1,294 | 21.5%  | +4.4           | 4,910  | 342      |
|          |          | 20    | 24,714 | 1,589 | 22.4%  | +5.3           | 5,187  | 360      |
|          | Semantic | 5     | 23,406 | 281   | 18.1%  | +1.0           | 3,941  | 298      |
|          |          | 10    | 23,647 | 522   | 18.9%  | +1.8           | 4,172  | 308      |
|          |          | 15    | 23,827 | 702   | 19.6%  | +2.5           | 4,349  | 311      |
|          |          | 20    | 23,886 | 761   | 19.8%  | +2.7           | 4,420  | 299      |
| Context  | Greedy   | 5     | 23,435 | 310   | 18.2%  | +1.1           | 3,860  | 408      |
|          |          | 10    | 23,672 | 547   | 19.0%  | +1.9           | 3,987  | 518      |
|          |          | 15    | 23,838 | 713   | 19.6%  | +2.5           | 4,164  | 507      |
|          |          | 20    | 23,999 | 874   | 20.1%  | +3.0           | 4,522  | 310      |
|          | Semantic | 5     | 23,266 | 141   | 17.6%  | +0.5           | 3,801  | 298      |
|          |          | 10    | 23,356 | 231   | 17.9%  | +0.8           | 3,881  | 308      |
|          |          | 15    | 23,398 | 273   | 18.1%  | +1.0           | 3,923  | 308      |
|          |          | 20    | 23,472 | 347   | 18.3%  | +1.2           | 3,992  | 313      |
| Both     | Greedy   | 5     | 23,159 | 34    | 17.2%  | +0.1           | 3,688  | 304      |
|          |          | 10    | 23,210 | 85    | 17.4%  | +0.3           | 3,727  | 316      |
|          |          | 15    | 23,224 | 99    | 17.5%  | +0.4           | 3,736  | 321      |
|          |          | 20    | 23,231 | 106   | 17.5%  | +0.4           | 3,762  | 302      |
|          | Semantic | 5     | 23,146 | 21    | 17.2%  | +0.1           | 3,681  | 298      |
|          |          | 10    | 23,160 | 35    | 17.2%  | +0.1           | 3,694  | 299      |
|          |          | 15    | 23,162 | 37    | 17.2%  | +0.1           | 3,695  | 300      |
|          |          | 20    | 23,175 | 50    | 17.3%  | +0.2           | 3,709  | 299      |
| Baseline |          |       | 23,125 | –     | 17.1%  | –              | 3,660  | 298      |

Table 10: Full augmented training-set statistics after quality filtering ( $Q(s) \geq 0.50$ ).

| Method   | Strategy | Shots | Aug.  | Full wojood  | $\Delta$     | Curras       | $\Delta$     | Dial.        | $\Delta$     | Lebanese     | $\Delta$     |
|----------|----------|-------|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Entity   | Greedy   | 5     | 615   | 86.08        | −0.40        | 75.19        | +4.84        | 72.67        | +2.49        | 40.98        | 40.98        |
|          |          | 10    | 1,007 | 85.76        | −0.72        | 75.56        | +5.21        | 74.18        | +4.00        | <b>50.00</b> | <b>50.00</b> |
|          |          | 15    | 1,294 | 85.93        | −0.55        | 75.20        | +4.85        | 73.78        | +3.60        | 48.78        | 48.78        |
|          |          | 20    | 1,589 | 85.95        | −0.53        | 75.03        | +4.68        | 73.87        | +3.69        | <b>50.00</b> | <b>50.00</b> |
|          | Semantic | 5     | 281   | 85.36        | −1.12        | 71.32        | +0.97        | 71.75        | +1.57        | 0.00         | —            |
|          |          | 10    | 522   | 86.05        | −0.43        | <b>76.93</b> | <b>+6.58</b> | <b>74.52</b> | <b>+4.34</b> | 6.90         | 6.90         |
|          |          | 15    | 702   | 86.02        | −0.46        | 76.37        | +6.02        | 73.99        | +3.81        | 4.76         | 4.76         |
|          |          | 20    | 761   | 85.47        | −1.01        | 72.73        | +2.38        | 71.42        | +1.24        | 0.00         | —            |
| Context  | Greedy   | 5     | 310   | <b>87.18</b> | <b>+0.70</b> | 75.38        | +5.03        | 73.26        | +3.08        | 16.67        | 16.67        |
|          |          | 10    | 547   | 87.03        | +0.55        | 76.34        | +5.99        | 73.24        | +3.06        | 36.04        | 36.04        |
|          |          | 15    | 713   | 87.14        | +0.66        | 76.88        | +6.53        | 73.72        | +3.54        | 19.05        | 19.05        |
|          |          | 20    | 874   | 75.17        | −11.31       | 73.88        | +3.53        | 72.56        | +2.38        | 0.00         | —            |
|          | Semantic | 5     | 141   | 87.02        | +0.54        | 74.61        | +4.26        | 73.02        | +2.84        | 0.00         | —            |
|          |          | 10    | 231   | 87.16        | +0.68        | 74.41        | +4.06        | 72.32        | +2.14        | 0.00         | —            |
|          |          | 15    | 273   | 87.17        | +0.69        | 75.09        | +4.74        | 73.20        | +3.02        | 0.00         | —            |
|          |          | 20    | 347   | 81.22        | −5.26        | 73.88        | +3.53        | 72.06        | +1.88        | 0.00         | —            |
| Both     | Greedy   | 5     | 34    | 85.91        | −0.57        | 74.26        | +3.91        | 73.36        | +3.18        | 0.00         | —            |
|          |          | 10    | 85    | 82.92        | −3.56        | 73.69        | +3.34        | 72.01        | +1.83        | 0.00         | —            |
|          |          | 15    | 99    | 86.11        | −0.37        | 73.06        | +2.71        | 71.11        | +0.93        | 0.00         | —            |
|          |          | 20    | 106   | 86.00        | −0.48        | 73.52        | +3.17        | 72.30        | +2.12        | 0.00         | —            |
|          | Semantic | 5     | 21    | 86.05        | −0.43        | 73.93        | +3.58        | 72.34        | +2.16        | 0.00         | —            |
|          |          | 10    | 35    | 85.93        | −0.55        | 74.18        | +3.83        | 73.64        | +3.46        | 0.00         | —            |
|          |          | 15    | 37    | 85.96        | −0.52        | 74.47        | +4.12        | 71.42        | +1.24        | 0.00         | —            |
|          |          | 20    | 50    | 82.98        | −3.50        | 73.72        | +3.37        | 72.37        | +2.19        | 0.00         | —            |
| Baseline |          |       |       | 86.48        | —            | 70.35        | —            | 70.18        | —            | 0.00         | —            |

Table 11: Complete NER results for all 24 configurations under flat augmentation. **Bold**: column maximum.

| Method                      | Tier  | Full Wojood |          | Curras       |              | Dialectal    |              | Lebanese†    |          |
|-----------------------------|-------|-------------|----------|--------------|--------------|--------------|--------------|--------------|----------|
|                             |       | F1          | $\Delta$ | F1           | $\Delta$     | F1           | $\Delta$     | F1           | $\Delta$ |
| Baseline                    | —     | 86.48       | —        | 70.35        | —            | 70.18        | —            | 0.00         | —        |
| Entity                      | top30 | 84.87       | -1.61    | 73.07        | +2.72        | 72.68        | +2.50        | 0.00         | —        |
| Entity                      | top60 | 80.06       | -6.42    | 73.59        | +3.24        | 71.74        | +1.56        | 0.00         | —        |
| Entity                      | all   | 86.75       | +0.27    | 72.83        | +2.48        | 71.42        | +1.24        | 2.47         | 2.47     |
| Context                     | top30 | 85.94       | -0.54    | 74.79        | +4.44        | 74.81        | +4.63        | 21.78        | 21.78    |
| Context                     | top60 | 85.77       | -0.71    | <b>76.05</b> | +5.70        | <b>74.85</b> | +4.67        | 36.97        | 36.97    |
| Context                     | all   | 86.57       | +0.09    | 75.30        | +4.95        | 73.71        | +3.53        | 37.17        | 37.17    |
| Both                        | all   | 80.08       | -6.40    | 73.59        | +3.24        | 71.71        | +1.53        | 0.00         | —        |
| Unified                     | top60 | 84.78       | -1.70    | 74.33        | +3.98        | 73.55        | +3.37        | <b>43.48</b> | 43.48    |
| Unified                     | all   | 84.61       | -1.87    | 74.33        | +3.98        | 74.53        | +4.35        | 34.43        | 34.43    |
| <b>Best (Context top60)</b> |       | 85.77       | -0.71    | <b>76.05</b> | <b>+5.70</b> | <b>74.85</b> | <b>+4.67</b> | 36.97        | 36.97    |

Table 12: Quality curriculum training by method and tier.  $\Delta$  vs. baseline. **Bold**: best F1 per column.

| Method  | Strategy | Shots | Top-30% tier |              |              |              | Top-60% tier |              |              |              |
|---------|----------|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|         |          |       | Full wjoood  | Curras       | Dial.        | Leb.         | Full wjoood  | Curras       | Dial.        | Leb.         |
| Entity  | Greedy   | 5     | 87.18        | 74.67        | 72.64        | —            | 87.19        | 75.08        | 72.61        | —            |
|         |          | 10    | 87.25        | 75.49        | 73.20        | —            | 87.14        | 74.94        | 73.08        | —            |
|         |          | 15    | 87.02        | 74.59        | 72.77        | —            | 87.11        | 74.63        | 72.73        | —            |
|         |          | 20    | <b>87.27</b> | <b>75.49</b> | 73.50        | —            | 87.24        | 74.85        | <b>74.51</b> | —            |
|         | Semantic | 5     | 86.90        | 74.50        | 71.54        | —            | 87.06        | <b>75.08</b> | 72.82        | —            |
|         |          | 10    | 87.15        | 74.73        | 72.68        | —            | <b>87.26</b> | 73.85        | 73.18        | —            |
|         |          | 15    | <b>87.27</b> | 74.87        | 72.71        | —            | 87.15        | 74.72        | 73.44        | —            |
|         |          | 20    | 87.08        | 73.53        | 71.99        | —            | 87.09        | 74.85        | 73.02        | —            |
| Context | Greedy   | 5     | 87.31        | 74.06        | 71.49        | 0.00         | 87.11        | 74.17        | 71.44        | 9.09         |
|         |          | 10    | 87.10        | 74.68        | 72.64        | <b>20.41</b> | 87.01        | 74.56        | 72.71        | <b>24.53</b> |
|         |          | 15    | <b>87.45</b> | <b>75.26</b> | <b>73.71</b> | 14.29        | <b>87.15</b> | 74.98        | <b>74.72</b> | 12.50        |
|         |          | 20    | 86.54        | 72.08        | 71.14        | 0.00         | 86.59        | 71.46        | 70.80        | 0.00         |
|         | Semantic | 5     | 87.10        | 73.77        | 71.69        | 0.00         | 87.08        | 74.22        | 71.94        | 0.00         |
|         |          | 10    | 86.70        | 72.82        | 71.65        | 0.00         | 86.97        | 73.76        | 71.25        | 0.00         |
|         |          | 15    | 87.36        | 73.84        | 71.65        | 0.00         | 87.01        | 74.04        | 72.06        | 0.00         |
|         |          | 20    | 86.77        | 72.88        | 71.27        | 8.00         | 86.80        | <b>75.06</b> | 72.29        | 10.00        |
| Both    | Greedy   | 5     | 86.57        | 73.41        | 69.57        | —            | 86.47        | 73.55        | 70.16        | —            |
|         |          | 10    | 86.84        | 74.73        | 73.97        | —            | 86.85        | 74.10        | 73.74        | —            |
|         |          | 15    | 87.13        | 73.53        | 72.32        | —            | 87.01        | <b>74.53</b> | 72.89        | —            |
|         |          | 20    | 87.05        | 72.07        | 70.30        | —            | <b>87.28</b> | 73.59        | 73.52        | —            |
|         | Semantic | 5     | 87.07        | 73.17        | 69.57        | —            | 87.15        | 73.43        | 71.90        | —            |
|         |          | 10    | 87.21        | 74.61        | 73.71        | —            | 87.05        | 73.60        | 72.12        | —            |
|         |          | 15    | <b>87.34</b> | <b>75.62</b> | <b>73.28</b> | —            | 87.12        | 73.88        | 72.56        | —            |
|         |          | 20    | 86.87        | 69.57        | 70.78        | —            | 87.13        | 68.91        | 71.22        | —            |

Table 13: Per-configuration curriculum NER results: top-30% and top-60% tiers. **Bold**: column maximum per tier. ‘—’ indicates no Lebanese entities (F1 = 0.00).

| Variant                                                 | Full wjoood  |              |       | Curras |              |       | Lebanese |              |       |
|---------------------------------------------------------|--------------|--------------|-------|--------|--------------|-------|----------|--------------|-------|
|                                                         | top30        | $Q \geq .50$ | all   | top30  | $Q \geq .50$ | all   | top30    | $Q \geq .50$ | all   |
| <i>(i) Dimension ablations and weight perturbations</i> |              |              |       |        |              |       |          |              |       |
| Original $Q(s)$                                         | 84.87        | 87.24        | 86.75 | 73.07  | 74.39        | 72.83 | 0.00     | <b>28.07</b> | 2.47  |
| –DA                                                     | 87.20        | 87.12        | 87.06 | 75.08  | 75.40        | 73.97 | 4.71     | 7.06         | 26.67 |
| –EI                                                     | 87.10        | 87.26        | 87.08 | 75.22  | 74.35        | 73.87 | 2.44     | 28.07        | 26.67 |
| –FL                                                     | 87.12        | 87.29        | 87.09 | 75.48  | 74.05        | 74.00 | 2.41     | 25.69        | 26.67 |
| –TU                                                     | <b>87.32</b> | 87.11        | 87.07 | 75.01  | 73.55        | 73.94 | 11.24    | 19.64        | 26.67 |
| Uniform 1/7                                             | 87.22        | 87.16        | 87.09 | 74.80  | 74.86        | 73.91 | 2.44     | 4.76         | 26.67 |
| DA-heavy 0.40                                           | 87.20        | 87.10        | 84.66 | 74.86  | 74.04        | 73.88 | 0.00     | 26.32        | 0.00  |
| <i>(ii) Threshold sensitivity (original weighting)</i>  |              |              |       |        |              |       |          |              |       |
| $Q \geq 0.50$                                           |              | 87.24        |       |        | 74.39        |       |          | 28.07        |       |
| $Q \geq 0.60$                                           |              | 87.33        |       |        | 73.90        |       |          | 0.00         |       |
| $Q \geq 0.70$                                           |              | 87.33        |       |        | 73.90        |       |          | 0.00         |       |

Table 14: Full  $Q(s)$  ablation. **Top panel**: dimension ablations and weight perturbations on the Entity-method pool, across test sets and training regimes. **Bottom panel**: threshold sensitivity under original weighting.

|                 | Original $Q(s)$ | Uniform 1/7 | DA +10% | DA –10% | EI +10% | FL +10% | TU +10% | Ablate DA | Ablate EI | Ablate FL | Ablate TU | DA-heavy 0.40 |
|-----------------|-----------------|-------------|---------|---------|---------|---------|---------|-----------|-----------|-----------|-----------|---------------|
| Original $Q(s)$ | 1.00            | 0.98        | 0.99    | 0.99    | 1.00    | 0.97    | 0.96    | 0.93      | 1.00      | 0.89      | 0.86      | 0.98          |
| Uniform 1/7     | 0.98            | 1.00        | 0.97    | 0.98    | 0.98    | 0.96    | 0.93    | 0.93      | 0.98      | 0.86      | 0.86      | 0.95          |
| DA +10%         | 0.99            | 0.97        | 1.00    | 0.97    | 0.99    | 0.96    | 0.95    | 0.89      | 0.99      | 0.89      | 0.86      | 1.00          |
| DA –10%         | 0.99            | 0.98        | 0.97    | 1.00    | 0.99    | 0.97    | 0.95    | 0.97      | 0.99      | 0.86      | 0.84      | 0.95          |
| EI +10%         | 1.00            | 0.98        | 0.99    | 0.99    | 1.00    | 0.97    | 0.96    | 0.93      | 1.00      | 0.89      | 0.86      | 0.98          |
| FL +10%         | 0.97            | 0.96        | 0.96    | 0.97    | 0.97    | 1.00    | 0.89    | 0.92      | 0.97      | 0.77      | 0.90      | 0.95          |
| TU +10%         | 0.96            | 0.93        | 0.95    | 0.95    | 0.96    | 0.89    | 1.00    | 0.90      | 0.96      | 0.93      | 0.69      | 0.93          |
| Ablate DA       | 0.93            | 0.93        | 0.89    | 0.97    | 0.93    | 0.92    | 0.90    | 1.00      | 0.93      | 0.78      | 0.77      | 0.86          |
| Ablate EI       | 1.00            | 0.98        | 0.99    | 0.99    | 1.00    | 0.97    | 0.96    | 0.93      | 1.00      | 0.89      | 0.86      | 0.98          |
| Ablate FL       | 0.89            | 0.86        | 0.89    | 0.86    | 0.89    | 0.77    | 0.93    | 0.78      | 0.89      | 1.00      | 0.64      | 0.89          |
| Ablate TU       | 0.86            | 0.86        | 0.86    | 0.84    | 0.86    | 0.90    | 0.69    | 0.77      | 0.86      | 0.64      | 1.00      | 0.86          |
| DA-heavy 0.40   | 0.98            | 0.95        | 1.00    | 0.95    | 0.98    | 0.95    | 0.93    | 0.86      | 0.98      | 0.89      | 0.86      | 1.00          |

Table 15: Spearman rank correlation between  $Q(s)$  variants on the full filtered pool.

| Variant       | $J_{\text{top30}}$ | $J_{\text{top60}}$ |
|---------------|--------------------|--------------------|
| Uniform 1/7   | 0.830              | 0.943              |
| DA +10%       | 0.933              | 0.958              |
| DA -10%       | 0.913              | 0.952              |
| EI +10%       | 1.000              | 1.000              |
| FL +10%       | 0.807              | 0.909              |
| TU +10%       | 0.737              | 0.878              |
| Ablate DA     | 0.800              | 0.864              |
| Ablate EI     | 0.999              | 1.000              |
| Ablate FL     | 0.605              | 0.835              |
| Ablate TU     | 0.552              | 0.799              |
| DA-heavy 0.40 | 0.878              | 0.922              |

Table 16: Jaccard overlap of top-30% and top-60% selected sets vs. original ranking.

| Variant         | Curras | Leb.  | Leb-Cur |
|-----------------|--------|-------|---------|
| Original $Q(s)$ | 0.561  | 0.574 | +0.013  |
| Uniform 1/7     | 0.473  | 0.482 | +0.009  |
| DA +10%         | 0.593  | 0.607 | +0.013  |
| DA -10%         | 0.520  | 0.533 | +0.013  |
| EI +10%         | 0.555  | 0.567 | +0.012  |
| FL +10%         | 0.569  | 0.579 | +0.010  |
| TU +10%         | 0.558  | 0.561 | +0.004  |
| Ablate DA       | 0.470  | 0.483 | +0.013  |
| Ablate EI       | 0.576  | 0.592 | +0.016  |
| Ablate FL       | 0.544  | 0.563 | +0.019  |
| Ablate TU       | 0.566  | 0.598 | +0.032  |
| DA-heavy 0.40   | 0.621  | 0.634 | +0.014  |

Table 17: Mean  $Q(s)$  per sub-corpus across variants. Positive Leb - Curras gap confirms no systematic under-scoring of Lebanese.

sentence 300000404:

**ORIG (Lebanese):**

ترفع السرية المصرفية عن كل السياسيين وشركاهن وعيلاتن  
ومدراهن العاميين وكبار موظفين الدولة ويلي تتقوا الشفافية،  
وينردو الأموال المنهوبة وينلغو الإحتكارات

**AUG (both-greedy-20):** السرية لماذا لا نرفع السرية

الانتخابات؟ عن كل الأوروبيين وعيلاتن ومدراهن والمسلمين  
السوريين وكبار موظفين الدولة قبل

The rewriting process often shifts dialectal Lebanese sentences toward formal MSA constructions. In one example, the original sentence containing the Levantine passive prefix ”-تن-” and colloquial verb forms such as ”تتقوا”, ”وينردو”, and ”وينلغو” is rewritten using the formal MSA interrogative scaffold ”لماذا؟! لا...”. This transformation removes key dialectal morphology and replaces it with syntactic patterns typical of written MSA. Several Curras examples exhibit even stronger drift toward newswire-style MSA. For instance, the colloquial nonsense rhyme ”وزير ثملة وحصانه” is rewritten as ”القومي أعلنت الحكومة صباح اليوم أن وزير البلاد” producing a fluent but entirely formal political-news sentence. Similarly,

the dialogue utterance

”في هذه الحالة” becomes ”الدكتور واضح انه صحته كويسة” in the dialogue utterance, introducing formal lexical choices and diacritics (e.g., kasra+shadda combinations) that are characteristic of written MSA rather than spoken dialect. Quantitatively, 30% of both-level augmentations contain at least one explicit MSA discourse marker, such as ”هذا السياق”, ”أعلنت”, ”إن”, ”حيث”, ”الذي”, while containing no dialectal markers at all. This explains the relatively low dialect-authenticity scores (DA  $\approx$  0.45–0.59) reported in §6.1.

**Entity corruption: loss and type flips.** Two label-level failure modes survive the entity-preservation filter. *Entity loss* occurs when at least one original entity is retained (thus satisfying the filter in Appendix A) while others disappear during rewriting.

For example, the 3-entity seed :

”الموظف : حسب الكتيب الفنان 10%”

(OCC, OCC, PERCENT) is rewritten as

”عندما رأيت الفنان على المسرح فوجئت بمواهبه في التمثيل”, which preserves only ”الفنان” while silently dropping both ”الموظف” and ”10%”. *Type flips* occur

when a preserved surface form changes semantic function during rewriting while its original gold label is copied unchanged, creating a mislabeled training example. Table 18 presents six such cases among the 106 both-level augmentations. For example, in the Curras dialogue ”يستقروا قصدون”, the token ”صباح” is a person name (PERS). After rewriting into ”حياتي”, the same surface form now means ”morning,” yet still retains the PERS label. These entity-faithfulness violations are especially common in small seed pools such as Lebanese, where substituted entities may violate type-level semantic compatibility (e.g., replacing a village-level GPE with a country). Importantly,  $Q(s)$  filters most of these violations effectively at the operational threshold  $Q(s) \geq 0.50$ .

**Generation artefacts and volume collapse.** A small subset of generations also contains non-linguistic artefacts that bypass the filtering pipeline. Specifically, 4/106 outputs (3.8%) contain embedded Latin-script tokens, e.g., ”البلاد في حالة هذه الأزمة”; 8/106 outputs (7.5%) reproduce quotation wrappers generated by the LLM itself; and 7/106 outputs (6.6%) collapse into short fragments of at most five tokens, such as ”صهانية، مالك، زلمة”.

None of these survive the  $Q(s) \geq 0.5$  filter, justifying  $Q(s)$  as a structural quality net. In contrast, curriculum filtering on top-30% or top-60% tiers successfully recovers performance to stable levels (86.4–87.5 F1) as shown in Table 13.

| Surface | ORIG tag | AUG tag  |
|---------|----------|----------|
| 3       | CARDINAL | ORDINAL  |
| 7       | ORDINAL  | CARDINAL |
| أربعة   | ORDINAL  | CARDINAL |
| صباح    | PERS     | TIME     |
| العربي  | NORP     | LANGUAGE |
| حماس    | ORG      | NORP     |

Table 18: Entity-type flips observed in 106 both-greedy-20 augmentations.

## I Generator Comparison: Qwen Full Data

We evaluate Qwen2.5-7B-Instruct on the three configurations that pinpoint LLaMA’s strongest results on Curras, Lebanese, and Full Wojood respectively. This subset is sufficient to test the portability of the augmentation strategies and the robustness of the quality pipeline without requiring an exhaustive 24-configuration sweep.

As illustrated in Table 19, Qwen generally preserves or slightly improves Full Wojood performance across configurations, while showing a more mixed behavior on dialect-specific subsets: Curras remains stable across generators, whereas Lebanese exhibits higher sensitivity to the generator choice, including configurations where Qwen collapses to zero.

These patterns align with the quality-controlled filtering dynamics defined by the pipeline.

| Config             | Generator | Full<br>Wojood | Curras       | Lebanese     | Dialectal    |
|--------------------|-----------|----------------|--------------|--------------|--------------|
| entity-semantic-10 | LLaMA     | 86.05          | <b>76.93</b> | 6.90         | 74.52        |
|                    | Qwen      | <b>87.22</b>   | 74.53        | 0.00         | 73.46        |
| entity-greedy-10   | LLaMA     | 85.76          | 75.56        | <b>50.00</b> | 74.18        |
|                    | Qwen      | <b>87.20</b>   | 75.39        | 24.30        | <b>74.38</b> |
| context-greedy-5   | LLaMA     | 87.18          | <b>75.38</b> | 16.67        | 73.26        |
|                    | Qwen      | <b>87.27</b>   | 75.27        | <b>24.30</b> | 74.32        |
| Baseline           | —         | 86.48          | 70.35        | 0.00         | 70.18        |

Table 19: Generator comparison: Qwen2.5-7B-Instruct vs. LLaMA-3.1-8B-Instruct on the three representative configurations evaluated.

| Dim.                                                     | Level    | Definition & Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Configuration-level diagnostics</i>                   |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| —                                                        | config   | <b>Dataset composition.</b> Sentence counts, token counts, and entity density per augmentation configuration. <i>Purpose: verifies that augmentation produces a sufficient volume of data and maintains realistic entity density relative to the base corpus.</i>                                                                                                                                                                                                       |
| —                                                        | config   | <b>Lexical diversity.</b> Type-token ratio (TTR), novelty score (fraction of augmented vocab absent from base), and Jaccard distance between augmented and base vocabularies. <i>Purpose: ensures augmented sentences introduce new vocabulary rather than paraphrasing existing sentences, which would add no new training signal.</i>                                                                                                                                 |
| —                                                        | config   | <b>Entity fidelity.</b> KL divergence between augmented and base entity-type distributions, average span length, and per-type frequency shifts. <i>Purpose: detects systematic entity-type drift — e.g. over-generation of PER at the expense of LOC — that could distort the NER model’s label priors.</i>                                                                                                                                                             |
| <i>Per-sentence dimensions (enter <math>Q(s)</math>)</i> |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DA                                                       | sentence | <b>Dialectal Authenticity.</b> Probability score from a character-ngram TF-IDF + Logistic Regression classifier (ngram_range=(2,4), max_features=30 000) trained on the base corpus to discriminate dialectal Lebanese/Curra sentences from MSA Wojood sentences. <i>Purpose: directly measures the core risk of dialectal augmentation — silent register drift toward MSA. A low DA score flags sentences that are fluent but useless for improving dialectal NER.</i> |
| EI                                                       | sentence | <b>IOB2 Validity.</b> Binary structural check: 1 if the tag sequence is fully IOB2-compliant, 0 otherwise. <i>Purpose: acts as a hard quality gate; a structurally invalid sentence cannot be used for NER training regardless of its linguistic quality.</i>                                                                                                                                                                                                           |
| SC                                                       | sentence | <b>Semantic Consistency.</b> Maximum cosine similarity between the augmented sentence and all original sentences in the same subcorpus, computed with LaBSE embeddings and shifted from $[-1, 1]$ to $[0, 1]$ . <i>Purpose: checks that the augmented sentence remains semantically grounded in the source domain.</i>                                                                                                                                                  |
| FL                                                       | sentence | <b>Fluency.</b> Mean per-token pseudo-log-likelihood (PLL) under CAMEL-Lab/bert-base-arabic-camelbert-mix, a dialect-aware Arabic MLM, MinMax-normalised within each configuration. <i>Purpose: estimates linguistic naturalness under a dialect-aware model.</i>                                                                                                                                                                                                       |
| TU                                                       | sentence | <b>Training Utility.</b> Proximity of the sentence’s UBC-NLP/MaRBERT CLS-embedding to the centroid of real dialectal base sentences, normalised so the closest sentence scores 1 and the furthest scores 0. <i>Purpose: estimates how likely the sentence is to improve generalisation to the real dialectal test distribution.</i>                                                                                                                                     |
| LN                                                       | sentence | <b>Lexical Novelty.</b> Fraction of content tokens (stop-words excluded) in the augmented sentence that are absent from the global base vocabulary. <i>Purpose: rewards sentences that introduce new lexical items, increasing the effective vocabulary coverage.</i>                                                                                                                                                                                                   |
| REP                                                      | sentence | <b>Repetition.</b> Unique bigram ratio $ \text{set}(\text{bigrams}) / \text{bigrams} $ , penalising degenerate or repetitive output. <i>Purpose: detects degenerate outputs typical of greedy decoding.</i>                                                                                                                                                                                                                                                             |
| <i>Supplementary (not in <math>Q(s)</math>)</i>          |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| —                                                        | sentence | <b>LLM-as-Judge.</b> Qwen2.5-7B-Instruct rates each sentence on five attributes (Fluency, Coherence, Label Correctness, Faithfulness, Dialectal Adequacy) on a 1–5 scale. <i>Purpose: provides a human-interpretable quality signal that cross-validates the automatic scores and surfaces failure modes not captured by the metric-based dimensions.</i>                                                                                                               |

Table 20: Full definitions of all quality pipeline dimensions. Dimensions marked *config-level* inform strategy selection only; *sentence-level* dimensions enter  $Q(s)$ .